Sprint 2 — Implementation and validation

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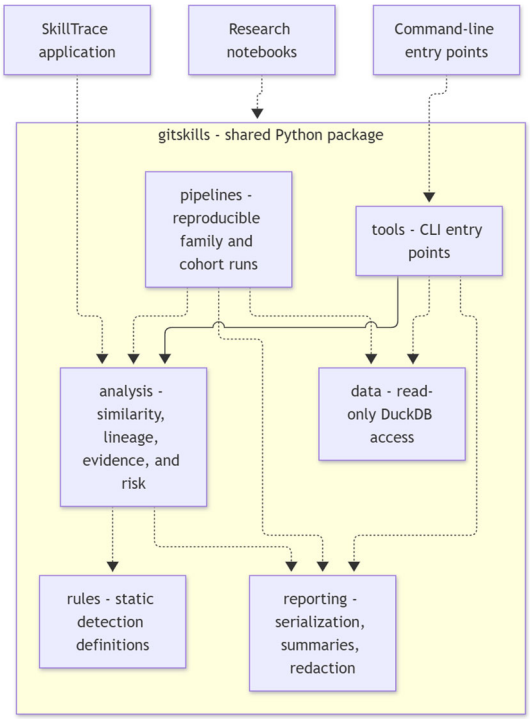
Due: October 28

The group must move from exploration to a defensible implementation. The Sprint 2 review must include:

- A working implementation of the core mining, extraction, similarity, classification, traceability, or visualization.
- Automated tests for important parsing, transformation, matching, or metric functions.
- A validation sample, manual annotation protocol, or other evaluation design.
- Preliminary results with reproducible figures or tables.
- At least one pull request reviewed by a different team member than the author.
- An updated research report containing the background, method, implementation, and preliminary results.
- A retrospective that records changes to the research question, method, scope, or interpretation.

Sprint 2 acceptance criteria: The core implementation runs end to end on the approved sample; the method is tested; the group can explain why the result should be trusted or where it may fail; and the preliminary analysis is supported by generated evidence rather than screenshots alone.

1) Our solution is built around a modular library shared across our visualization applications, research notebooks, and command-line tools.



We identify potentially related SKILL.md artifacts using three similarity criteria:

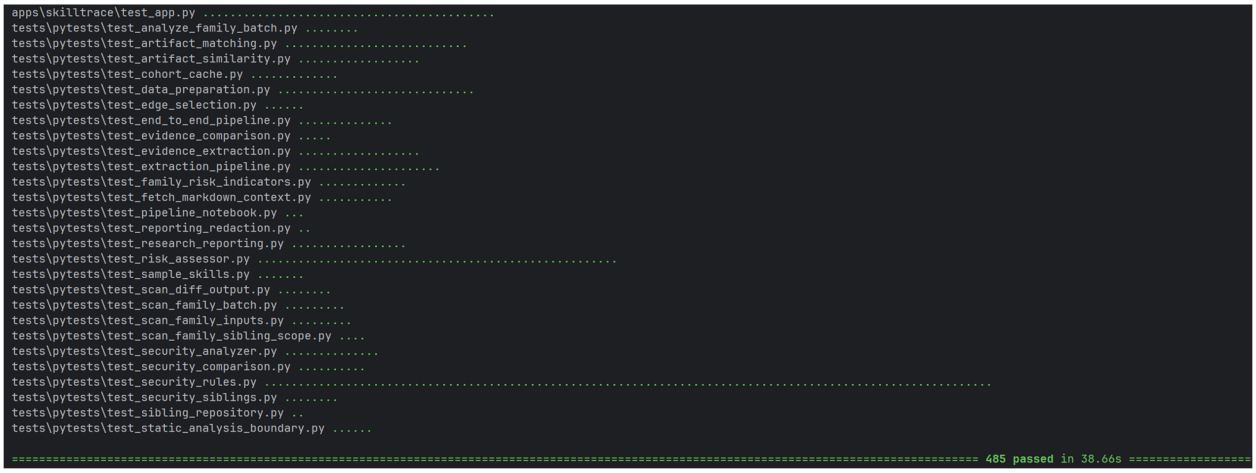
- At least 5 matching 5-token shingles.
- At least 80% containment in one direction.
- A Jaccard similarity of at least 30%.

All three criteria must be satisfied for two artifacts to be considered related. We then use chronological information to infer potential lineage between matching artifacts.

Once related artifacts are identified, we apply static detection rules to identify security-sensitive behavior, extract contextual evidence, and use risk indicators to assess potential security concerns. By comparing related artifacts, we can identify capabilities introduced or removed through modification or reuse.

Additional details about our implementation and design decisions are available in ARCHITECTURE.md and DESIGN.md.

2) We maintain an extensive test suite for all aspects of our library, applications, tools, and notebooks. All tests, including edge cases, pass successfully:



3) We define a manual_annotation_protocol.md and provide validation samples, annotation templates, and manual review files under the end-to-end pipeline results folder:

   For similarity analysis:
   results/sprint2/manual-review/relationship-pairs/relationship_validation_sample.csv
   results/sprint2/manual-review/relationship-pairs/relationship_annotations_template.csv
   results/sprint2/manual-review/relationship-pairs/annotations.csv

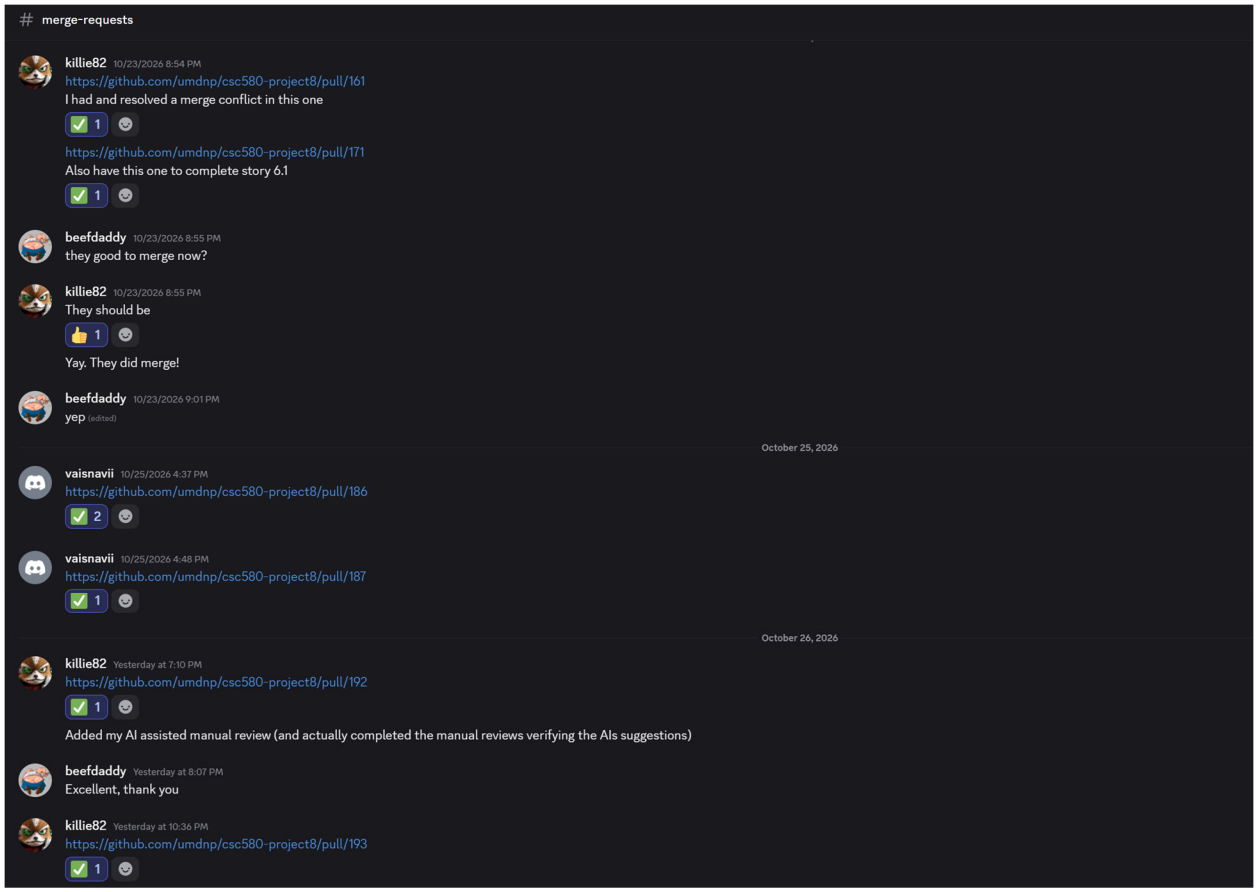
   For rule & risk detection:
   results/sprint2/manual-review/directed-pairs/validation_sample.csv
   results/sprint2/manual-review/directed-pairs/annotations_template.csv
   results/sprint2/manual-review/directed-pairs/annotations.csv

4) Our end-to-end pipeline is managed through a Jupyter Notebook available here:
   notebooks/end_to_end_pipeline/end_to_end_pipeline.ipynb
   (with preliminary run results)

   All tables are stored as csv files under sprint2/review, and figures can be found under sprint2/figures.

5) Our public GitHub repository is available at csc580-project8, and our Sprint 2 backlog is managed in the Team 8 GitHub Project. The main branch is protected by rules that restrict deletions, require a pull request before merging, and block force pushes.

   All pull requests are communicated on our project Discord server:



6) Our report containing background, method, implementation, and preliminary results can be found in report/final.pdf.

7) Our Sprint 2 retrospective, including what we learned and how it affected the backlog, is available in docs/retrospective/sprint-2.md

8) Our security threat model is documented in THREAT_MODEL.md. The primary risks, limitations, and threats to validity identified during Sprint 2 are documented in THREATS_TO_VALIDITY.md.